*Seminar Presentation*
*on*

# Title: A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company

Paper: Rodrigues, Šírová & Mugurusi (2022), IFAC-PapersOnLine 55(10), pp. 2773-2778


***by***

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# INTRODUCTION

- A small Czech company (about **400 workers**) makes parts for the car industry.
- **Copper rod** is its most critical spare part (found by ABC analysis). No rod in stock means the **production line stops**.
- Problem: spare parts are costly and suppliers are often late.
- **Three suppliers** can sell the rod, and **three managers** must agree on one.
- Until now the supplier was chosen by experience only.
- **Aim of the paper:** a clear scoring model to choose the supplier, solved with **AHP** and **ANP**.

| Supplier | Located in | What the paper says |
|---|---|---|
| **Supplier 1** | Germany | Best quality, highest price, farthest away |
| **Supplier 2** | Czech Republic (Central Bohemia) | Low price, fastest delivery |
| **Supplier 3** | Czech Republic (North Bohemia) | Low price, weakest quality |

**Maintenance manager** (4 yrs)
Leads the purchase, negotiates price and delivery

**Production manager** (5 yrs)
A line stop costs more than any delivery charge

**Quality manager** (10 yrs)
Zero-defect policy: quality above price and time

Price **€230 to €350** per rod | Delivery **3 to 7 days** | Bad parts **6.67% to 15%** (2018) | Spend about **€25,300** a year

Source: Rodrigues, Šírová & Mugurusi (2022), Sections 3 and 4.

# WHAT THE PAPER DID

1. Chose **6 attributes** in **2 groups** (Fig. 1).
2. Each manager compared **two items at a time** on Saaty's **1 to 9 scale**.
   1 = equal, 9 = extremely more
3. The three answers were combined by **geometric mean** into **8 tables** (paper Tables 2 to 9).
4. **SuperDecisions** software turned each table into weights and checked consistency.
5. Solved the problem two ways:
   **AHP:** tree, both groups equally important
   **ANP:** network, groups affect each other

**Result:** AHP picks **Supplier 2**, ANP picks **Supplier 1**. The paper recommends the ANP answer.

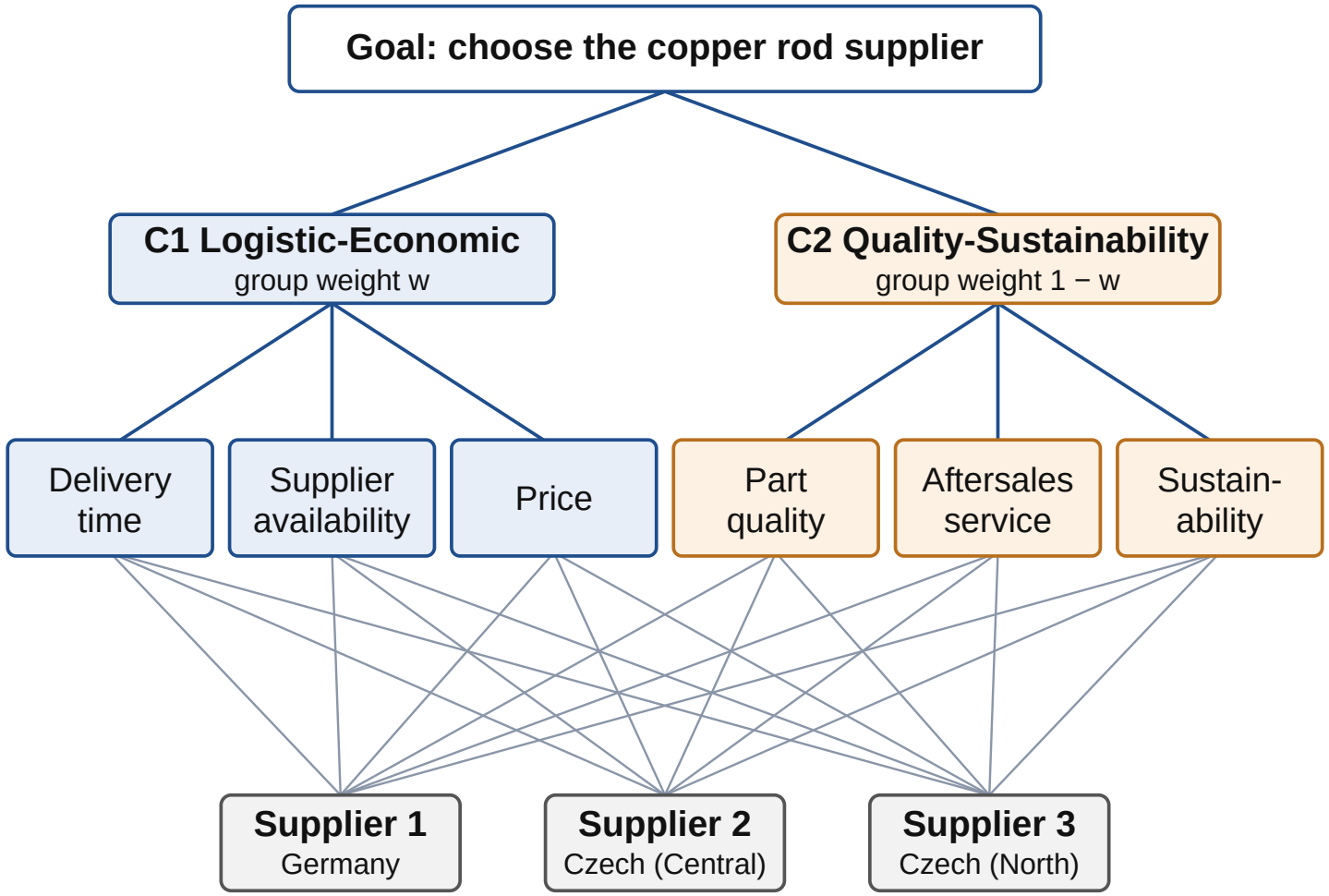



Fig. 1: Decision tree of the paper (redrawn from Rodrigues et al., 2022, Fig. 1)

Source: Rodrigues et al. (2022), Sections 4 and 5.

# GAPS IDENTIFIED IN THE PAPER

| Gap | What I found from the paper's own tables |
|---|---|
| **1. Two methods, two winners** | I rebuilt both results exactly. **ANP equals plain AHP** with the quality group made **5 times** more important (w = 1/6). |
| **2. No sensitivity check** | The winner changes at **w = 0.442**, very close to the w = 0.5 used by AHP (Fig. 2). |
| **3. Answers not consistent** | **4 of 8 tables** are above Saaty's 5% limit (8.25%, 5.16%, 6.24%, 5.16%). |
| **4. Measured values given as opinions** | Supplier 1 costs only **1.52 times** more (€350 vs €230), but its price score is **7.3 times** lower. |
| **5. One fixed ranking** | No check of how sure the winner is when answers vary a little. |
| **6. No cost view** | The ranking ignores money lost on bad rods and line stops. |

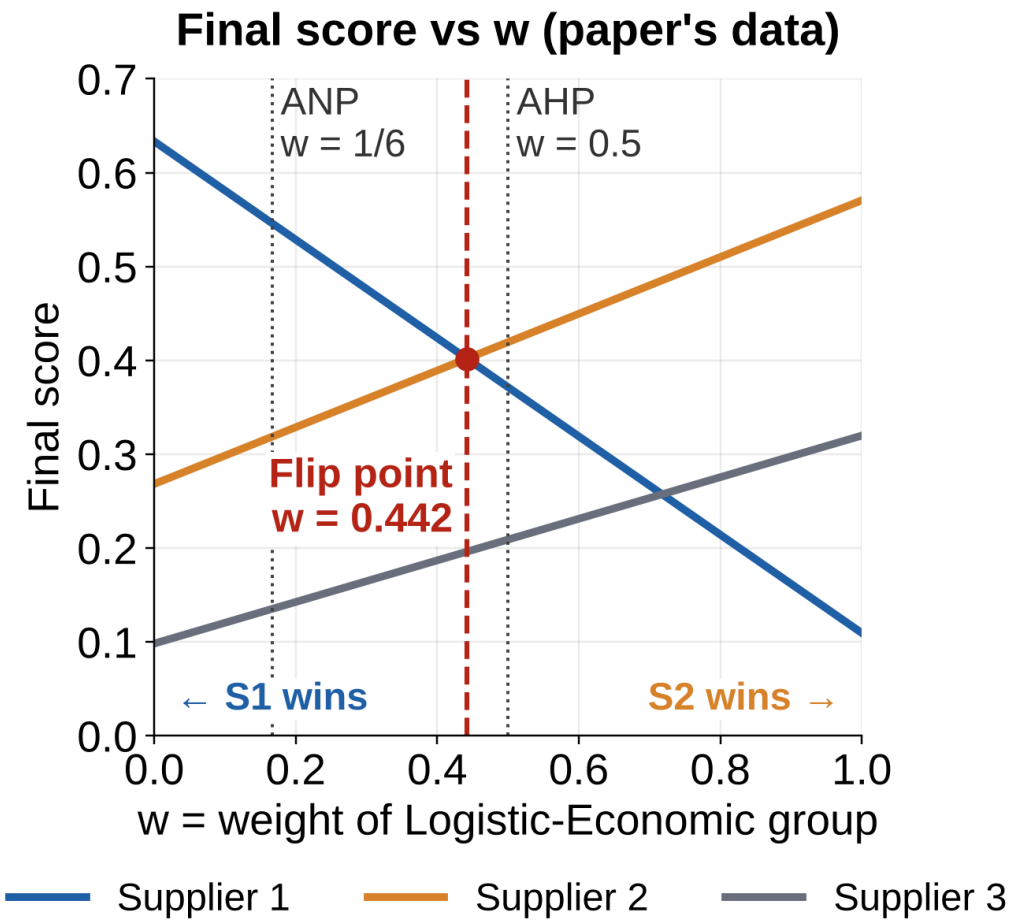



Fig. 2: Same tables, only w changes. Below 0.442 Supplier 1 wins, above it Supplier 2.

Source: own recalculation from Rodrigues et al. (2022), Tables 2 to 12.

# MY PROPOSED IMPROVEMENT

**Same input values as the paper.** Only the way they are used is improved.

1. **Re-check doubtful answers** (gap 3)
   In each failing table, move one answer by one step, so all 8 tables pass.

2. **Use real numbers where they exist** (gap 4)
   Price, delivery days and defect rate: score = (1/x) ÷ Σ(1/x), which keeps the true ratio.

3. **Show the flip point** (gaps 1, 2)
   Managers see exactly which group weight changes the winner.

4. **Test how sure the winner is** (gap 5)
   5,000 trials: each answer moved by ±1 step, w picked at random from 0.1 to 0.9.

5. **Add a 2-year cost per buying plan** (gap 6)
   Only S1, only S2, only S3, and a 50/50 split of S1 and S2.

### Answers to re-check

| Table | Answer | Change | CR after |
|---|---|---|---|
| T2 | Price vs Availability | 1/3 → 1/2 | **2.36%** |
| T3 | Aftersales vs Sustain. | 2 → 1 | **0.00%** |
| T5 | Quality: S1 vs S2 | 3 → 2 | **1.36%** |
| T7 | Aftersales: S2 vs S3 | 2 → 1 | **0.00%** |

### 2-year cost of a plan

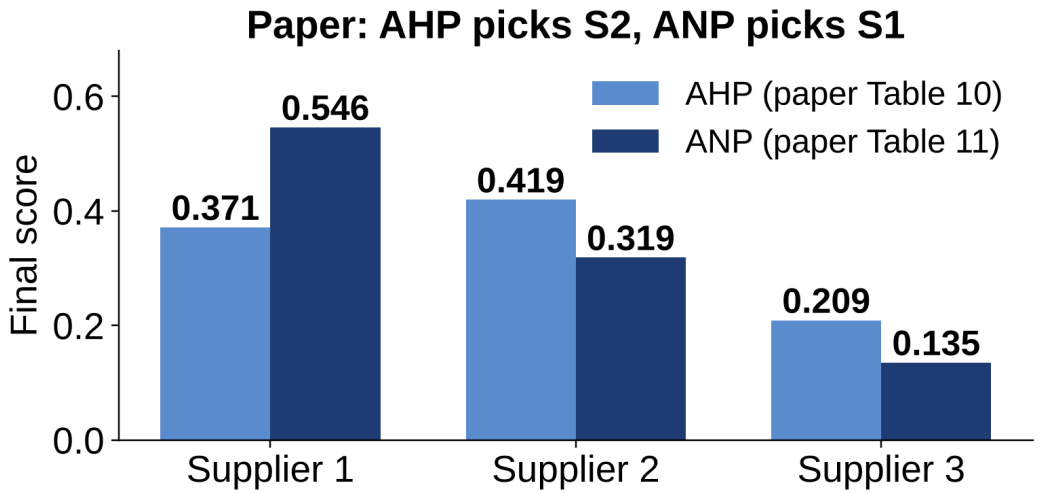
**Cost** = rods × price
+ bad rods × €150
+ line-stop hours × €300

Price, delivery and defect values from the paper's ranges; €150 and €300 are estimates the company can replace.

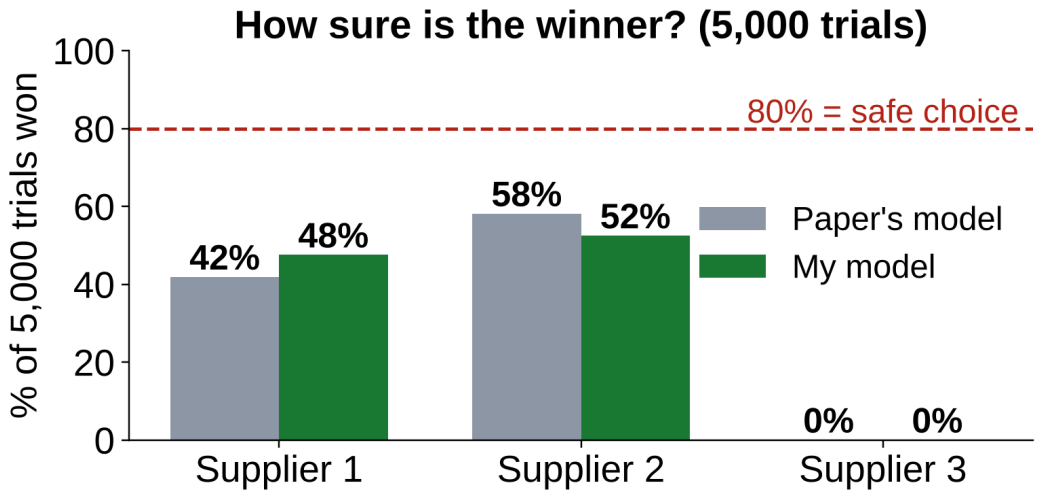
All steps are built into a simple web tool that shows the input values and the working (node server.js, then open localhost:8080).

# RESULTS: PAPER'S MODEL vs MY MODEL

## Paper's model




## My model




| | Paper's model | My model |
|---|---|---|
| **Winner** | Depends on the method. **No reason given.** | S2 0.374, S1 0.367 at w = 0.5: a **close call**, flips at w = 0.486 |
| **Consistency** | **4 of 8** tables above 5% | **8 of 8** tables within 5% |
| **Price gap** | Scored as **7.3 times** | True **1.52 times** kept |
| **How sure?** | Not checked | S2 wins **52%**, S1 **48%**, S3 **0%** of trials |
| **2-year cost** | Not shown | S2 **€56.9k**, split **€72.2k**, S3 **€71.6k**, S1 **€87.5k** |

Source: left, paper Tables 10 and 11; right, my web tool using the same paper Tables 2 to 9.

# CONCLUSIONS AND FUTURE WORK

## Conclusions

1. The AHP and ANP winners differ because of **one group weight**, not because of the network method.
2. Small, honest changes in the answers make S1 and S2 swap places. A **single ranking hides this**.
3. Fair scoring (consistent answers and real numbers) keeps **S2 just ahead**.
4. **Cost** turns the close call into a clear choice: S1 pays off only if one bad rod costs over **€5,250**.

## Future work

- Fuzzy AHP for answers the managers are unsure about.
- Give managers different weights by experience.
- Use the company's real yearly data in place of estimates.
- Find the best order split by optimisation.
- Link the tool to the company's ERP data.

**Recommendation:** drop Supplier 3 and buy from Supplier 2. If the quality risk is high, split the order 50/50 between Supplier 1 and Supplier 2.